

Rear Drive Axle and Differential



The rear axle assembly consists of the following:

- Integral-type housing hypoid gear design (center of the pinion set below the centerline of the ring gear)
- Hypoid differential ring gear and pinion consists of an 8.8-inch ring gear and an overhung drive pinion that is supported by 2 opposed tapered roller bearings
- Axle housing assembly that consists of a cast center section with 2 steel tube assemblies and a stamped differential housing cover
- Differential pinion shaft is retained by a threaded differential pinion shaft lock bolt attached to the differential case
- Differential case is mounted in the rear axle housing between 2 opposing differential bearings that are retained in the rear axle housing by 2 removable bearing caps

The rear drive axle receives rotational torque from the driveshaft. The direction of the rotation determines movement of the vehicle. The rotation enters the rear drive axle through the drive pinion flange, turning the drive pinion and rotating the ring gear. The ring gear is bolted to the differential case. The differential case houses the pinion shaft upon which the differential pinion gears ride. The pinion gears are in mesh with the differential side gears to which the axle shafts are splined. It is the unique arrangement of the differential that allows the axle shafts to be driven at different speed when needed, such as when driving around corners. A Traction-Lok differential is also available that introduces a series of clutch plates and discs behind the differential side gears to deliver as much torque as possible to each axle shaft when the traction at the rear wheels is not equal. The Traction-Lok differential requires a friction modifier that must be installed with the axle lubricant for the plates and discs to be functional.